

Prediction and Clustering of Life Expectancy: Supervised and Unsupervised Learning Approaches

Abstract

This study explores both supervised and unsupervised learning models to analyze and understand the determinants and patterns of life expectancy across countries. We employ various regression models to predict life expectancy using GDP, Adult Mortality and infant deaths as input features. Additionally, we apply clustering algorithms to detect patterns among countries based on their life expectancy relative to GDP and Adult Mortality. Our findings indicate that multivariate regression outperforms simple models in prediction, while k-Means and hierarchical clustering reveal distinct clusters aligned with economic and health-related factors.

1. Introduction

Life expectancy is a critical metric used to evaluate the overall health and quality of life in countries around the world. This report aims to explore supervised and unsupervised learning techniques to understand the factors that influence life expectancy and to identify natural clusters among countries based on selected health and economic indicators.

2. Methodology

Part I: Supervised Learning to Predict Life Expectancy

Exploratory Data Analysis

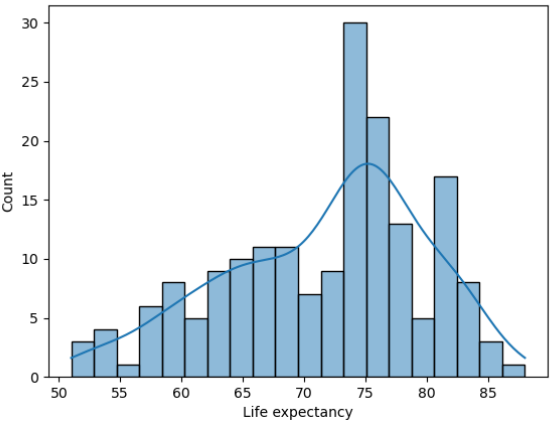


Figure 1

Distribution of life expectancy values across countries

This plot shows a roughly normal distribution with a peak around 70-75 years

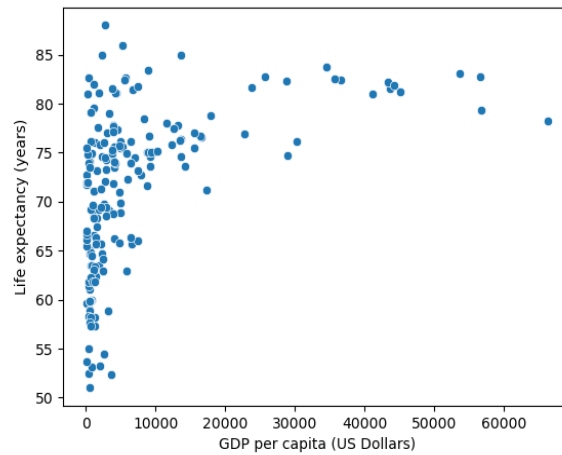


Figure 2

A Scatterplot of GDP vs Life Expectancy

This scatterplot reveals a non-linear, upward pattern. We notice that wealthier countries have higher life expectancy.

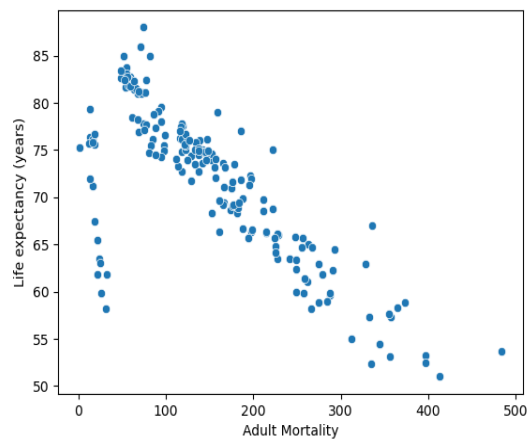


Figure 3

A Scatterplot of Adult Mortality vs Life expectancy

This scatterplot reveals a strong negative linear relationship. This makes intuitive sense since adult mortality impacts the average lifespan.

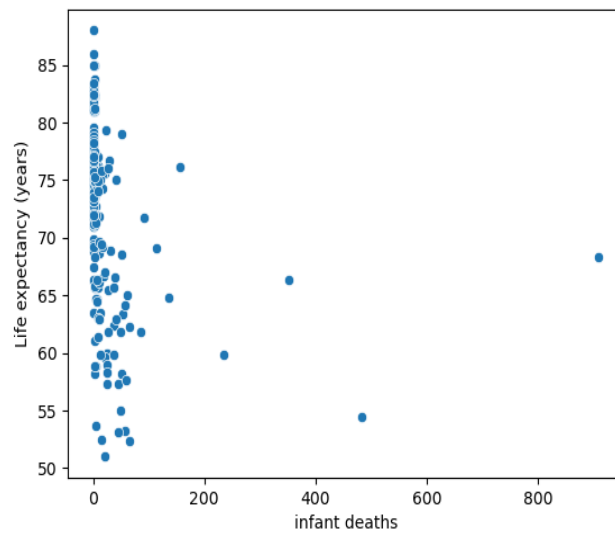


Figure 4

A Scatterplot of Infant Deaths vs Life Expectancy

Infant deaths also appear inversely related to life expectancy but with a more scattered trend

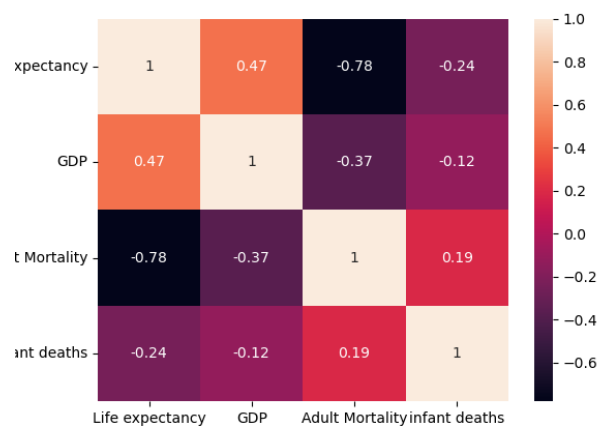


Figure 5

Correlation matrix

Correlation matrix shows strongest correlation of life expectancy with adult mortality (negative)

A. Using A Simple Linear Regression Model

We trained simple linear regression models using the following features individually:

Feature	MAE	MSE	RMSE	R ²
GDP	6.18	58.34	7.63	0.15
Adult Mortality	3.61	27.82	5.27	0.59
Infant Deaths	6.80	67.93	8.24	0.01

Conclusion:

Adult Mortality yields the best single-feature prediction (highest R² and lowest errors), suggesting a strong inverse relationship with life expectancy.



Figure 6
Linear Regression plot of GDP vs Life Expectancy

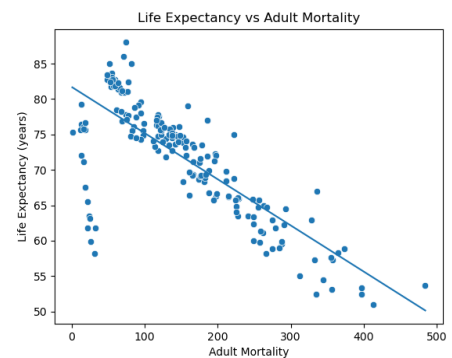


Figure 7
Linear Regression plot of Adult Mortality vs Life Expectancy

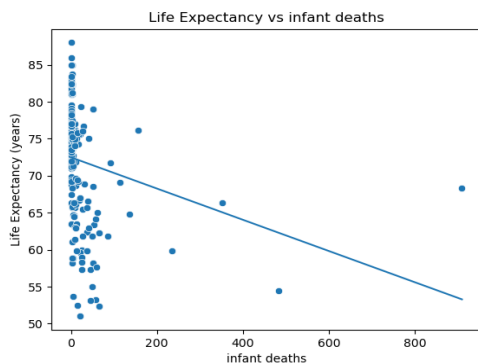


Figure 8
Linear Regression plot of Infant deaths vs Life Expectancy

B. Polynomial Regression (GDP vs Life Expectancy)

To address the non-linear relationship, we fit polynomial models of degrees 2 to 5:

Degree	MAE	MSE	RMSE	R ²
2	5.51	50.90	7.13	0.25
3	5.46	49.00	7.00	0.28
4	5.45	47.96	6.92	0.30
5	5.30	46.49	6.81	0.32

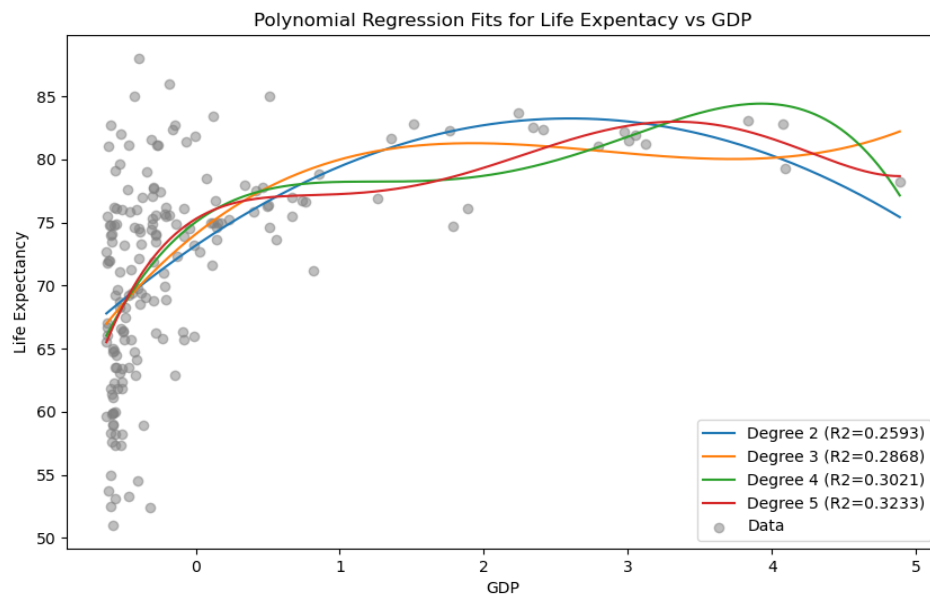


Figure 9

Polynomial Regression of degrees 2-5 (GDP vs Life Expectancy)

Conclusion:

Degree 5 polynomial gives the best result, showing the highest R² and lowest errors

C. Multiple Feature Models

Using GDP, Adult Mortality and Infant deaths as predictors, we compared:

Model	MAE	MSE	RMSE	R ²
Linear Regression	3.56	26.20	5.11	0.61
Ridge Regression	3.57	26.02	5.11	0.61

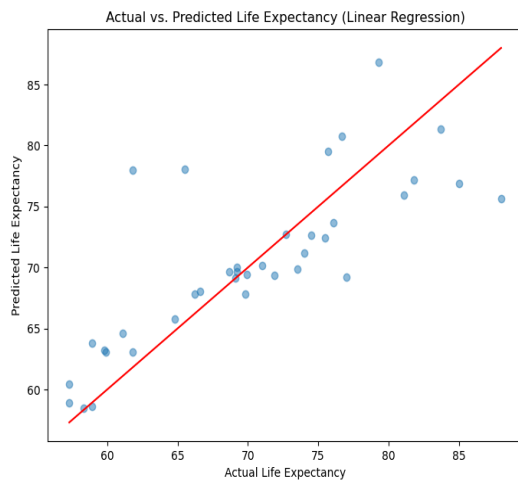


Figure 10

Linear Regression with Multiple input variables

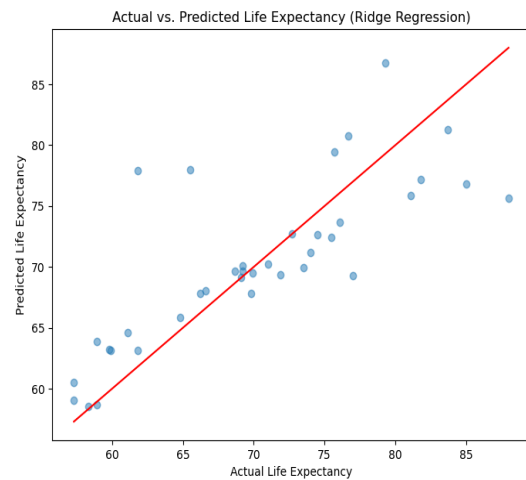


Figure 11

Ridge Regression with Multiple input variables

Conclusion:

The Linear and Ridge models perform similarly. But we see that adding more features improves prediction accuracy rather than using a single variable model.

D. Best Model Summary

The multiple linear regression model offers the best prediction performance based on R² and error metrics. Among simple models, adult mortality is best single predictor. For non-linear models, 5th-degree polynomial regression on GDP is optimal

Part II: Unsupervised Learning to Cluster Countries

A. K-Means Clustering

Clustering by GDP and Life Expectancy

- Elbow Method: Optimal Clusters = 3
- Silhouette Score = 0.54
- Davies-Bouldin Score: 0.56

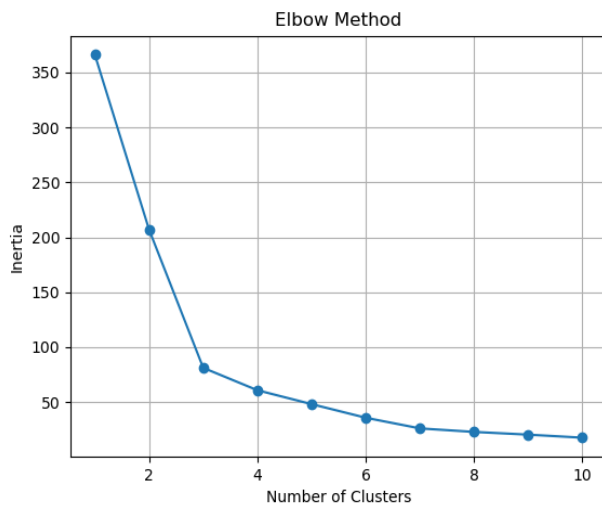


Figure 12

*Elbow Method showing Optimal number of Clusters
GDP and Life expectancy*

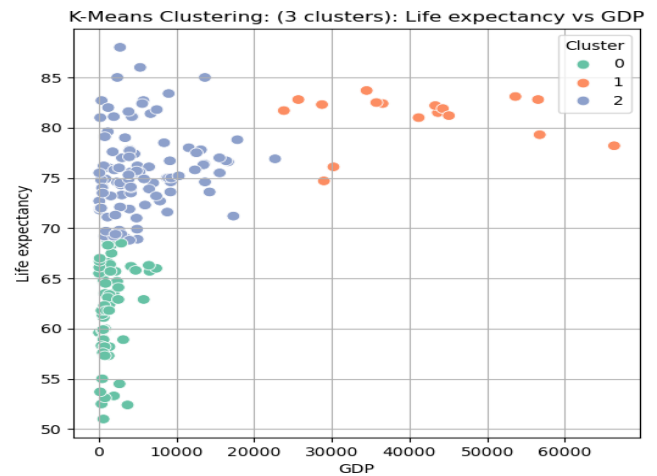


Figure 13

K-Means Clustering: GDP vs Life Expectancy

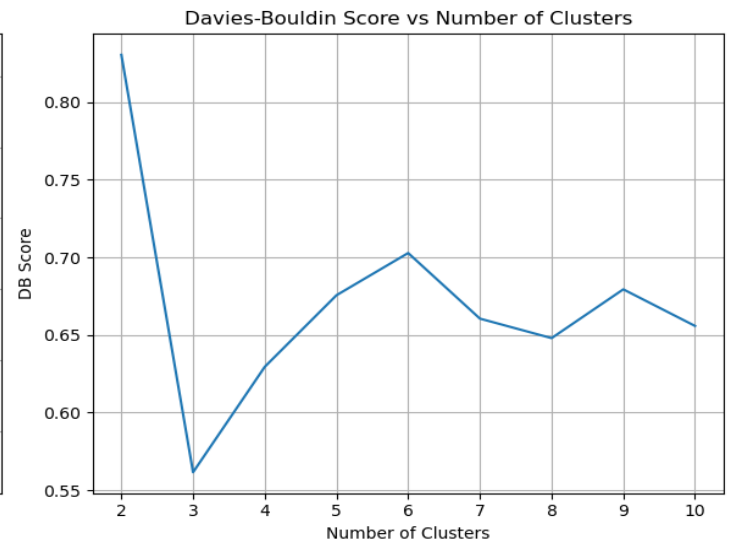
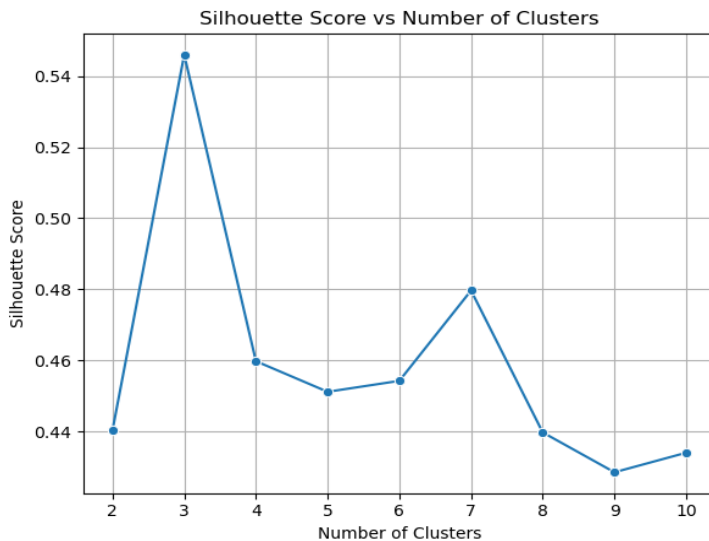


Figure 14

Silhouette and Davies-Bouldin Scores

Interpretation:

The Silhouette Score indicates that the clusters are moderately well-separated and the Davies-Bouldin Score suggest that the clusters have a relatively reasonable separation, though there is room for improvement in cluster distinctness.

Clustering by Adult Mortality and Life Expectancy

- Elbow Method: Optimal Cluster = 3
- Silhouette Score: 0.43
- Davies-Bouldin Score: 0.79

K-Means Clustering: (3 clusters): Life expectancy vs Adult Mortality

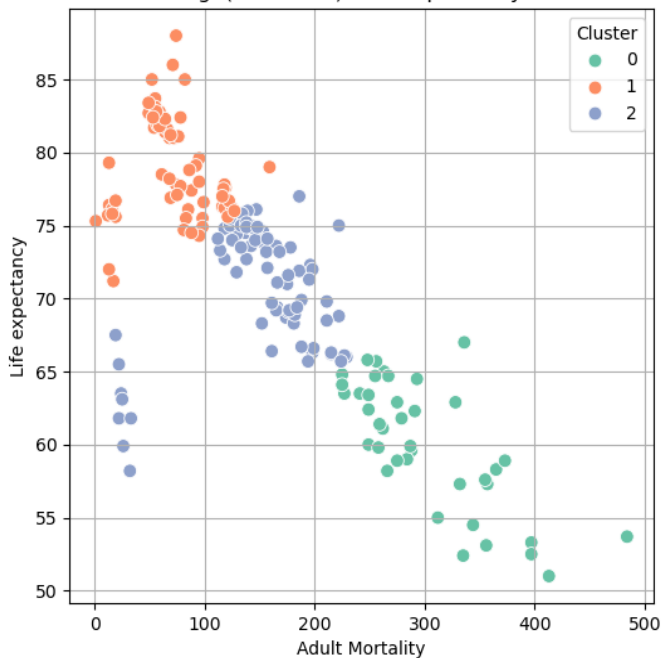


Figure 15

*Elbow Method showing Optimal number of Clusters
Adult Mortality vs Life Expectancy*

Elbow Method

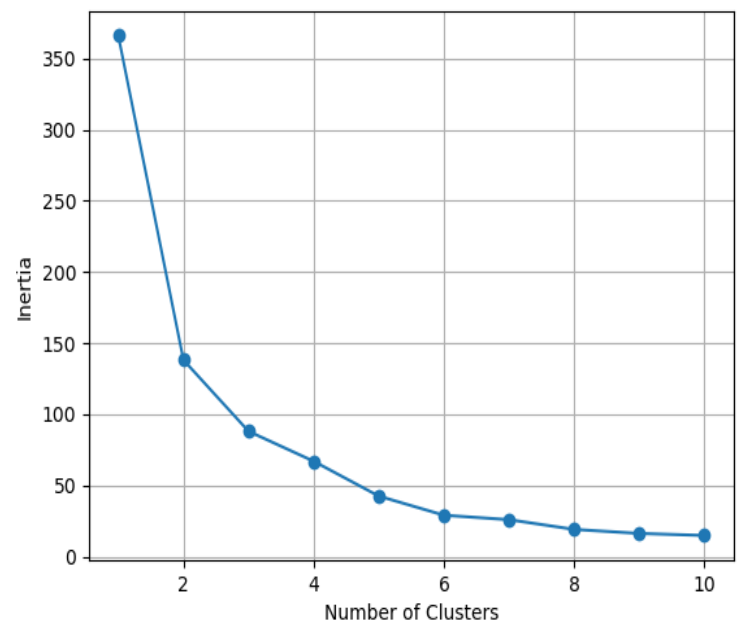


Figure 16

*K-Means Clustering:
Adult Mortality vs Life Expectancy*

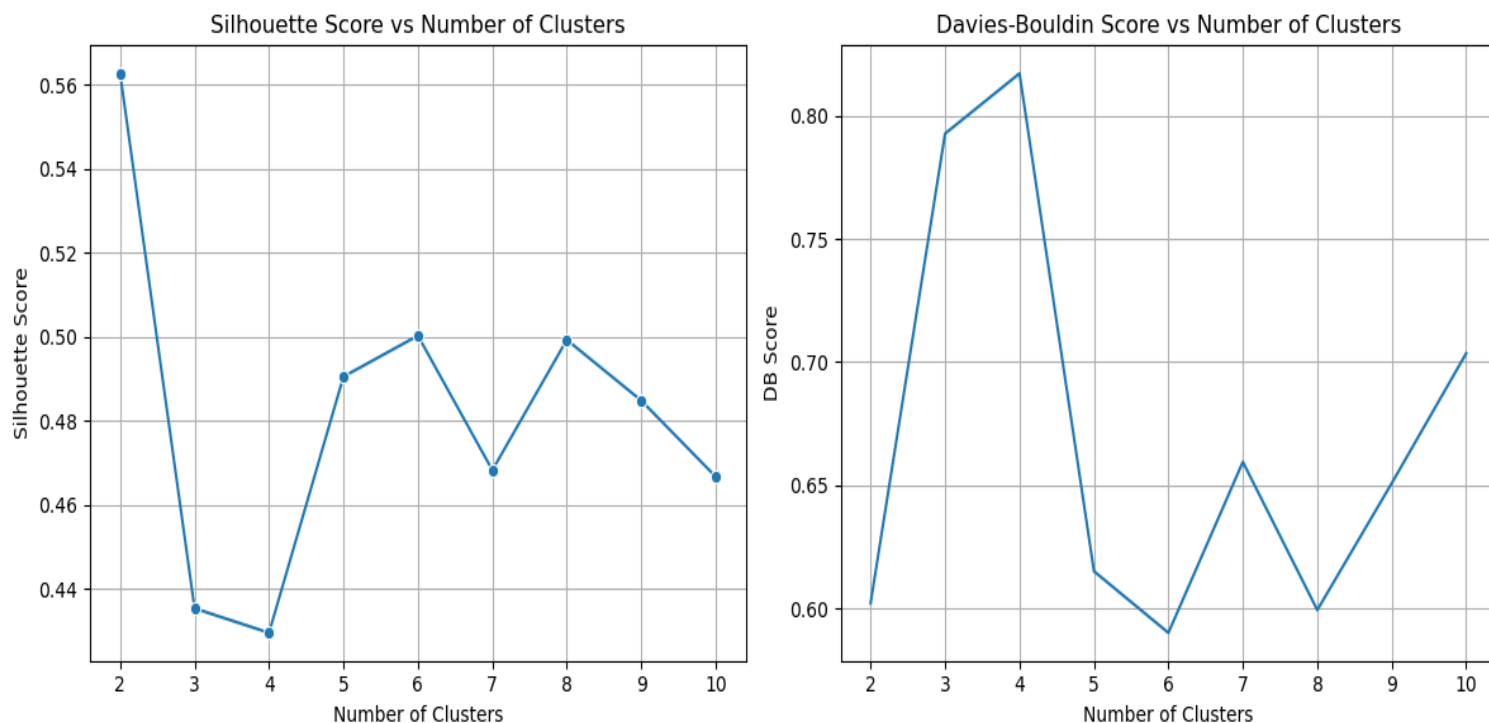


Figure 17

Silhouette and Davies-Bouldin Scores

Conclusion: K-Means clustering using GDP and Life Expectancy produced better- defined clusters based on both evaluation metrics.

- Higher silhouette score (0.53 vs 0.43)
- Lower Davies-Bouldin score (0.56 vs 0.79)

This suggest that GDP may be a more informative variable for clustering that GDP in this dataset, naturally segmenting countries into distinct economic and health categories- such as low income with low life expectancy, and high income with high life expectancy.

These cluster are meaningful because they reflect systemic influences on healthcare, nutrition and sanitation. The utility here lies in being able to identify countries facing similar developmental and public health constraints, which can inform policy intervention

N/B: A lower silhouette score (closer to 0) indicates that clusters are not well-separated and that many data points lie close to the boundary between clusters while a higher Davies-Bouldin score means the clusters are less distinct.

B. Comparison of Clustering Algorithms

All clustering methods below were applied using GDP and Life Expectancy as input features for consistency:

Agglomerative Clustering

Silhouette Score: 0.56

Davies-Bouldin Score: 0.44

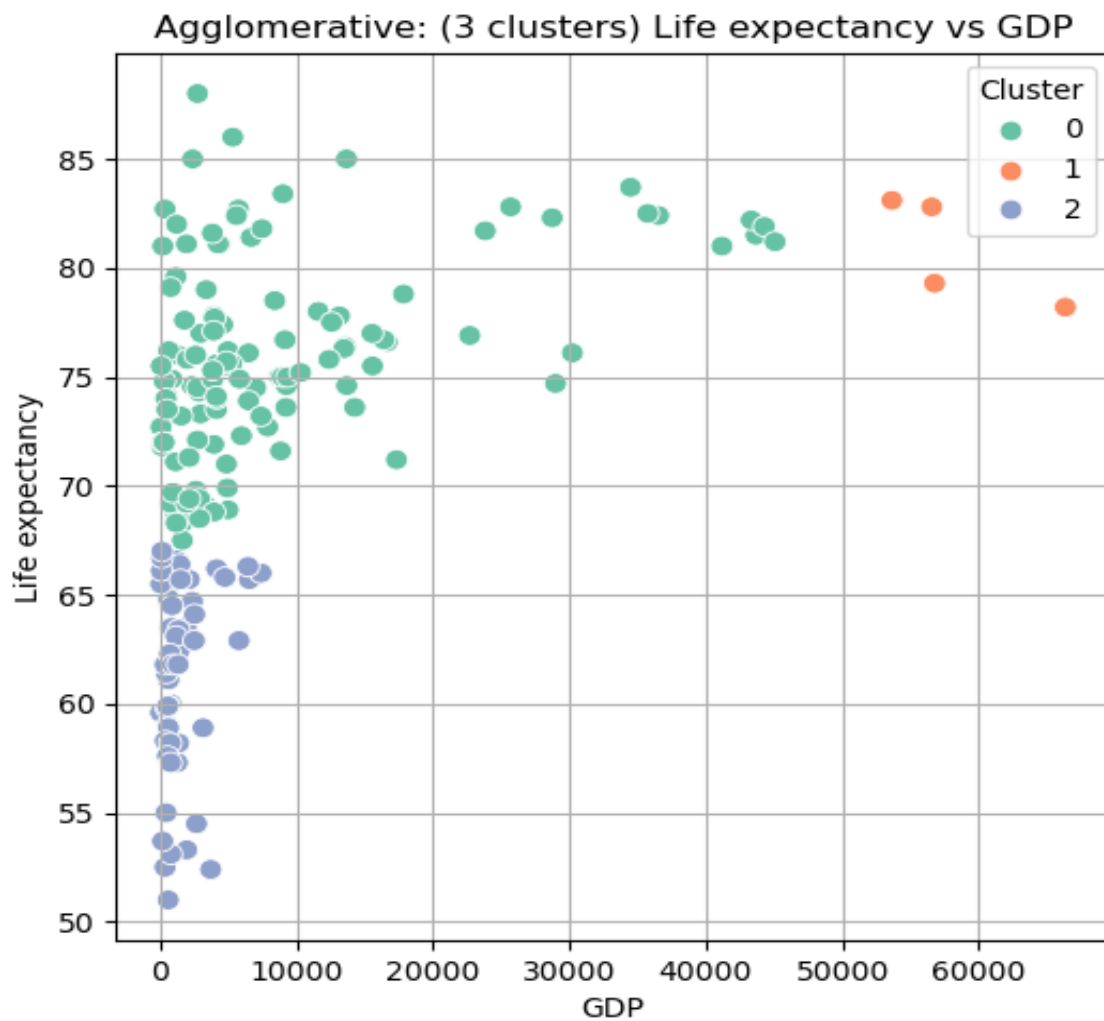


Figure 18

Agglomerative Clustering of GDP vs Life Expectancy

DBSCAN

Silhouette Score: 0.27

Davies-Bouldin Score: 0.61

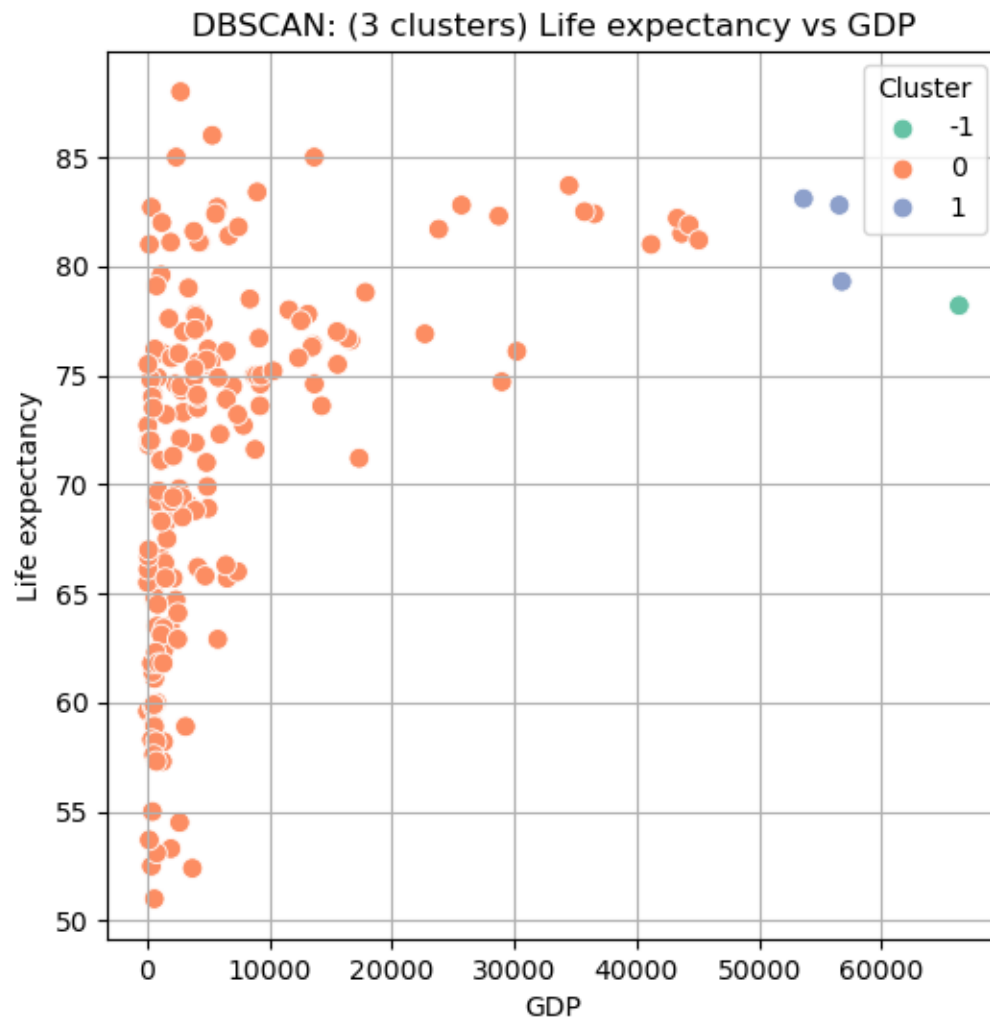


Figure 18

DBSCAN Clustering of GDP vs Life Expectancy

Model Comparison and interpretation

Best Performing Clustering Model:

Model	Silhouette Score	Davies-Bouldin Score
Agglomerative	0.56	0.44
K-Means	0.54	0.56
DBSCAN	0.27	0.61

- **Agglomerative Clustering** yields the clusters Silhouette Score (0.56) and the lowest Davies-Bouldin Score (0.44), suggesting it produces the best-defined and most compact clusters for this dataset.
- **K-Means** performs closely, but slightly worse, especially in terms of the Davies-Bouldin Score (which is higher, indicating more overlap between clusters).
- **DBSCAN** performs significantly worse, with a low Silhouette Score and high Davies-Bouldin Score, likely due to its sensitivity to density and shape of the clusters

Cluster Membership Consistency:

The countries are not generally assigned to the same clusters across all models:

- **Agglomerative Clustering and K-Means** may produce similar groupings if the data structure is mostly spherical and hierarchical in nature, but exact assignments will vary depending on how each algorithm defines separation.
- **DBSCAN**, on the other hand, is fundamentally different as it detects arbitrary-shaped clusters based on density. Many countries may be assigned to different clusters or even marked as noise (outliers) in DBSCAN that would otherwise belong to defined clusters in K-Means or Agglomerative.

Final Conclusion

This study applied both supervised and unsupervised learning methods to investigate the factors influencing life expectancy across countries and to identify meaningful clusters based on health and economic indicators. In the supervised analysis, multiple linear regression provided the most accurate predictions, with Adult Mortality emerging as the most influential variable. Polynomial regression was useful in capturing non-linear trends, particularly involving GDP.

In the unsupervised analysis, Agglomerative Clustering yielded the most distinct groupings especially when using GDP and Life expectancy as features. These clusters aligned well with known global health disparities. K-Means clustering offered moderately effective result while the DBSCAN struggled to define meaningful distinctions due to its sensitivity to data density.

Overall, the combination of predictive modeling and clustering techniques offer a valuable framework for understanding global health patterns. These insights can inform policy decisions by highlighting key predictors of life expectancy and revealing structural patterns in a country's health data